DEPARTMENT OF CS
UNIVERSITY OF ENGINEERING & TECHNOLOGY
PESHAWAR

DATA SCIENCE MINI PROJECT GUIDELINES

End-to-End Data Science Pipeline Implementation

Course: Data Science

Project Type: Group Project

Group Size: 3–4 Students

---

1. Project Overview

The purpose of this mini-project is to provide students with practical experience in executing a complete Data Science workflow, starting from problem identification and data collection to machine learning model development and business insight generation.

Students will work collaboratively to solve a real-world problem using data-driven methodologies and modern machine learning techniques.

---

2. Learning Objectives

Upon successful completion of this project, students will be able to:

- Define and formulate a real-world data science problem.
- Collect and preprocess datasets from public sources.
- Perform exploratory data analysis (EDA).
- Apply feature engineering and feature selection techniques.
- Build and evaluate machine learning models.
- Optimize model performance using hyperparameter tuning.
- Present findings and recommendations to technical and non-technical audiences.

---

3. Project Requirements

Dataset Requirements

Students must select a dataset that satisfies the following conditions:

- Minimum 1,000 records.
- Minimum 10 features/attributes.
- Publicly available dataset.
- Suitable for:
  - Classification
  - Regression
  - Clustering

Recommended Sources

- Kaggle
- UCI Machine Learning Repository
- OpenML
- Government Open Data Portals

---

4. Suggested Project Domains

Students may select projects from domains such as:

- Healthcare
- Finance
- Banking
- Education
- E-commerce
- Sports Analytics
- Environmental Monitoring
- Transportation
- Agriculture
- Social Media Analytics

Example Topics

- Customer Churn Prediction

- Loan Default Prediction
- Diabetes Prediction
- Student Performance Analysis
- Air Quality Prediction
- Sales Forecasting
- Employee Attrition Prediction

---

5. Project Phases

Phase 1: Problem Definition and Exploratory Data Analysis (EDA)

Objectives

Understand the problem and explore the dataset.

Tasks

1. Problem Definition

Clearly define:

- Problem statement
- Business objective
- Machine learning task:
  - Classification
  - Regression
  - Clustering

2. Data Collection

- Obtain dataset from approved sources.
- Explain data source and dataset characteristics.

3. Data Quality Assessment

Identify:

- Missing values
- Duplicate records
- Outliers
- Incorrect data types
- Inconsistent values

4. Exploratory Data Analysis

Perform visualizations using:

- Matplotlib
- Seaborn

Recommended analyses:

- Histograms
- Boxplots
- Correlation heatmaps
- Scatter plots
- Count plots
- Pair plots

Deliverables

- Jupyter Notebook
- Dataset description
- Data dictionary
- EDA report (2–3 pages)

Phase 2: Feature Engineering and Baseline Modeling

Objectives

Transform raw data into machine-learning-ready features.

Tasks

A. Feature Engineering

Apply appropriate techniques such as:

Feature Transformation

- Log Transformation
- Standard Scaling
- Min-Max Scaling
- Robust Scaling

Feature Encoding

- Label Encoding

- One-Hot Encoding
- Ordinal Encoding

Feature Creation

Create meaningful new features where applicable.

Examples:

- BMI from height and weight
- Total spending from individual purchases
- Age groups from age

B. Feature Selection

Apply at least TWO feature selection methods.

Filter Methods

- Correlation Analysis
- Chi-Square Test

Wrapper Methods

- Recursive Feature Elimination (RFE)
- Forward Feature Selection
- Backward Elimination

Embedded Methods

- LASSO Regression
- Tree-Based Feature Importance

Compare selected features and justify the final feature set.

C. Baseline Model Development

Build at least one baseline model.

Examples:

Classification

- Logistic Regression
- Naïve Bayes

Regression

- Linear Regression

Evaluate using appropriate metrics.

Deliverables

- Cleaned dataset
- Feature engineering notebook
- Feature selection results
- Baseline model evaluation

Phase 3: Advanced Modeling and Optimization

Objectives

Improve model performance and validate results.

Tasks

A. Advanced Models

Implement at least TWO advanced machine learning models.

Examples:

Classification

- Random Forest
- Support Vector Machine (SVM)
- Gradient Boosting
- XGBoost

Regression

- Random Forest Regressor
- Gradient Boosting Regressor
- Support Vector Regressor

B. Hyperparameter Tuning

Apply one of the following:

- GridSearchCV

- RandomizedSearchCV

Show:

- Parameters tested
- Best parameter combination
- Performance improvement

C. Dimensionality Reduction (Bonus)

Apply:

- Principal Component Analysis (PCA)

or

- Singular Value Decomposition (SVD)

Compare:

- Performance before reduction
- Performance after reduction

D. Model Evaluation

Use appropriate evaluation metrics.

Classification Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- ROC Curve
- AUC Score

Regression Metrics

- MAE
- MSE
- RMSE
- R^2 Score

Include visualizations wherever applicable.

6. Final Report Structure

The report should not exceed 10 pages.

Recommended structure:

1. Introduction
2. Problem Statement
3. Dataset Description
4. Data Preprocessing
5. Exploratory Data Analysis
6. Feature Engineering
7. Feature Selection
8. Model Development
9. Results and Discussion
10. Conclusions and Recommendations
11. References

7. Presentation Guidelines

Duration

8–10 Minutes

Presentation Structure

1. Project Title
2. Problem Statement
3. Dataset Overview
4. EDA Findings
5. Feature Engineering
6. Feature Selection
7. Machine Learning Models
8. Evaluation Results
9. Business Insights

10. Conclusion

All team members must participate in the presentation.

8. Submission Requirements

Submit the following:

Source Code

- Jupyter Notebook (.ipynb)
- Python scripts (.py)

Documentation

- Final Report (PDF)

Presentation

- PowerPoint Slides (PPT/PDF)

Repository

- GitHub Repository Link

Repository must include:

- Source code
- Dataset link
- README file
- Project documentation

9. Evaluation Rubric

Criterion	Weight
Problem Definition & EDA	20%
Feature Engineering & Selection	20%
Model Development	25%
Model Evaluation & Optimization	20%
Report & Presentation	15%

Total = 100%

---

10. Academic Integrity

- All work must be original.
- Properly cite datasets and external resources.
- Plagiarism will result in disciplinary action.
- Students must be able to explain all submitted code and results during evaluation.

Recommended Project Workflow

Problem Definition ↓ Dataset Collection ↓ Data Cleaning ↓ EDA ↓ Feature Engineering ↓ Feature Selection ↓ Baseline Model ↓ Advanced Models ↓ Hyperparameter Tuning ↓ PCA/SVD (Optional) ↓ Evaluation ↓ Business Insights ↓ Final Report & Presentation